Trace–Answer Compatibility Emerges at Depth and Shapes Answer Prediction in Diffusion Language Models

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Abstract

We study what trace–answer readouts measure in LLaDA-8B and Dream-7B, using Qwen3-8B, Llama-3.1-8B, and Dream's AR initialization family, Qwen2.5-7B, as autoregressive references. A decoder trained to distinguish original and crossed pairings, without correctness labels, reaches within-question AUC 0.919 and 0.897 at deep blocks. On GSM8K, much of this signal tracks numeric consistency between the trace and final answer. We then test conditional prediction by re-masking short answers and patching whole late answer-slot states. These patches recover 66% and 80% of the trace-conditioned answer-likelihood gaps on GSM8K, and 71% and 86% on BIG-Bench Hard multiple-choice and Yes/No questions. LLaDA decoding and patching replicate under an independent generation seed. Deep pre-answer controls are near null in all five models, but answer-position effects emerge later in the diffusion models. The tested decoder direction has no measurable patch effect; a cross-fitted 64-dimensional trace-difference subspace retains 70% and 84% of the whole-state effect. Together, these analyses characterize late trace–answer readouts and quantify how interventions on late hidden states change conditional answer prediction.

1 Introduction

Diffusion language models generate by iteratively reconstructing masked tokens rather than predicting strictly left-to-right [Austin et al., 2021, Lou et al., 2024]. No fixed position summarizes a prefix, and token confidence need not track solution quality. Nevertheless, linear readouts [Alain and Bengio, 2018] can decode whether a generation is correct [Yadav et al., 2026]. This success leaves an interpretive problem: correctness AUROC can reflect question difficulty, answer form, trace properties, or trace–answer compatibility. We need to distinguish these signals to interpret the readout.

We examine this ambiguity in the natively trained LLaDA-8B [Nie et al., 2025] and AR-initialized Dream-7B [Ye et al., 2025], with Qwen3-8B and Llama-3.1-8B as AR references. Comparing correct and wrong attempts to the same question, with answer-only and first-block baselines, reveals a late correctness-associated direction in both diffusion models on GSM8K. Readouts transfer between GSM8K and MATH at late blocks; the AR references transfer from their first block.

To examine the readout's content, we cross non-answer traces and final answers. Pairing-dependent scores motivate a decoder trained on pairing provenance, without correctness targets. Deep all-position AUC reaches 0.919 in LLaDA and 0.897 in Dream; the decoder also transfers across formats. Shallow contrast directions are weak or unaligned; splice-specific cues remain possible.

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We then test conditional prediction by inserting the same correct answer into both traces, re-masking it, and patching its predictive states [Heimersheim and Nanda, 2024]. Deep whole-state patches recover 66% and 80% of the mean trace-conditioned likelihood gaps. LLaDA patching replicates under a second seed; both models show late effects on short non-numeric BBH answers. Intermediate-number edits and step shuffling also change the readout and likelihood.

Deep pre-answer controls are near null in all models; the AR comparison instead identifies a timing difference. AR answer-position effects are present from the first block (7–15% of the gap), while diffusion effects remain at or below 3% through half depth.

Our contributions are: (1) within-question and factorial readout analyses, including pairing decoding without correctness targets; (2) objective-aligned whole-state interventions with seed and non-numeric controls; and (3) AR references, including Dream's initialization family, that delimit these depth differences.

Combining probe score, agreement, and confidence has an observed advantage of only 0.011 over majority voting ($p = 0.14$), and probe-guided allocation matches random (Appendix F). Trace source does not certify process validity, and the comparisons do not isolate diffusion training as the cause of the depth difference or locate reasoning computation. The features and source-token paths that construct late states remain open.

2 Related Work

Linear probes show whether information can be read from a hidden state [Alain and Bengio, 2018]. Probe success alone does not establish that the model uses that information, computes it at the probed layer, or represents it independently of input cues [Hewitt and Liang, 2019, Belinkov, 2021]. Text-only and first-block baselines help test whether simple input cues explain success.

TrajSelector, CLUE, LiLaVe, and DUCHESS use latent readouts to improve best-of-N selection over confidence or self-consistency [Yu et al., 2025, Liang et al., 2025, Piotrowski et al., 2025, Jiang et al., 2025]. This use makes the comparison unit important: a selector ranks attempts to the *same* problem, while pooled AUROC includes cross-problem pairs. Within-question AUC measures this ordering and better predicts selection gain in the examined settings (Appendix B).

In base-to-diffusion comparisons, Kong et al. [2026] find task-dependent circuit changes, including earlier processing on globally constrained tasks. This cautions against assigning a universal depth profile to diffusion training. For correctness readouts, Yadav et al. [2026] find decodable information in LLaDA and Dream despite weak token confidence. Their protocol uses one generation per prompt and reports pooled AUROC. A strong prompt-only control indicates a substantial difficulty signal. Our protocol bridge reproduces this separation (Appendix E); multiple draws allow within-question comparisons, surface controls, and trace–answer recombination.

Causal mediation and activation patching localize information use in AR transformers [Vig et al., 2020, Meng et al., 2023, Heimersheim and Nanda, 2024]. Adapting these tests to bidirectional models requires matching the predictive objective: AR scoring predicts answers left-to-right, while diffusion scoring re-masks them in the completed solution. We use recombination to expose a readout interaction and bidirectional whole-state patches to test conditional answer scoring. Component patches provide finer location tests, but their effects are non-additive. The complete circuit and natural mediation by the decoded feature remain unresolved.

3 Experimental Setup

Models and data. We compare the natively trained masked-diffusion model LLaDA-8B-Instruct and the AR-initialized diffusion model Dream-7B, with Qwen3-8B [Yang et al., 2025] and Llama-3.1-8B-Instruct [Grattafiori et al., 2024] as similarly sized autoregressive references (Table 1). Dream was initialized from Qwen2.5-7B [Ye et al., 2025]. We also evaluate Qwen2.5-7B-Instruct [Yang et al., 2024] from that family, with the same 28 blocks and hidden width of 3584. Other post-training differences prevent this closer comparison from isolating the training objective.

GSM8K [Cobbe et al., 2021] is the primary task, with MATH [Hendrycks et al., 2021] as a second numeric-reasoning task. Layer analyses use $n = 150$ questions and $K = 6$ independent generations

Table 1: Models in the main comparison.

Model	Class	Blocks	Hidden size
Qwen3-8B	autoregressive	36	4096
Llama-3.1-8B-Instruct	autoregressive	32	4096
LLaDA-8B-Instruct	masked diffusion	32	4096
Dream-7B-Instruct	AR-initialized diffusion	28	3584
Qwen2.5-7B-Instruct	autoregressive	28	3584

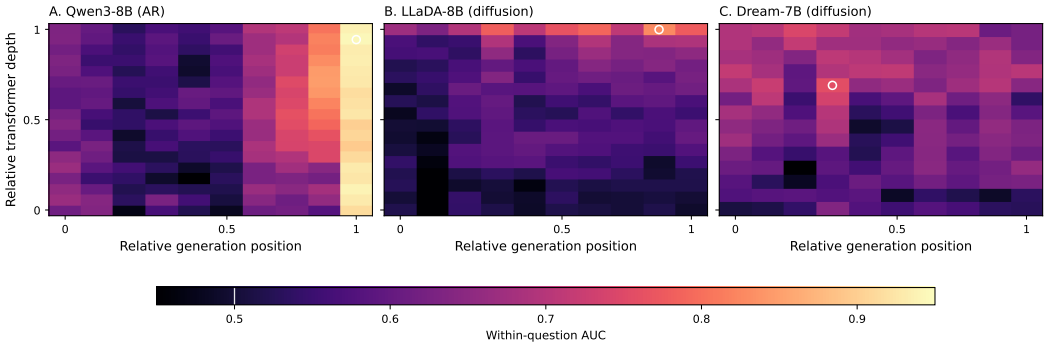


Figure 1: Primary-run layer-by-position within-question AUC. Qwen3 shows a terminal-position stripe from shallow depth; LLaDA's signal emerges in deep blocks across several positions, while Dream is weaker and less terminally localized. Open circles mark each model's highest-AUC cell; all panels share one color scale.

per model at temperature 0.9. GSM8K uses the first 640 training-split examples. AR sampling uses nucleus sampling with $p = 0.95$; diffusion decoding uses 128 denoising steps over a 256-token region. Correctness is exact numeric match. Dream's one-position logit shift, inherited from its AR initialization, is applied during both generation and scoring. To test non-numeric answers, we use 358 questions from eight BIG-Bench Hard tasks [Suzgun et al., 2022], with six samples per question and no targeted resampling (Appendix K).

Measurement. We report *within-question* AUC to measure how well the readout orders attempts to the *same* question: only correct–incorrect pairs within that question contribute. Cross-validation groups generations by question. All probe and causal bootstraps treat the question as the independent unit and keep its variants together (Appendix A).

The main *all-position* readout averages the means of ten equal-width relative-position bins over the generated region. For diffusion models, this is the full 256-token canvas, including end-of-sequence filler after the answer. We fit an L2-regularized logistic probe at each sampled block, using standardized states saved for every second block.

We compare these scores with an *answer-only* baseline, using seven features of the parsed answer string, and a *first-block* readout, using the output of transformer block 0 rather than the raw embedding. We also compute a training-free *re-masked conditional answer likelihood*: the mean log-probability assigned to the detected answer span after masking it, given the rest of the solution. AR models instead use teacher-forced mean answer-token log-probability.

4 Correctness Readouts Across Depth on GSM8K

We first ask *when* correct and wrong attempts become linearly distinguishable. At fixed GSM8K endpoints (Table 2), all-position within-question AUC rises from 0.451 at block 0 to 0.826 at block 30 in LLaDA and from 0.556 to 0.800 in Dream. Qwen3 starts at 0.820 and gains little. The layer-by-position map (Fig. 1) shows how this pattern varies across tokens: Qwen3 has a terminal-position stripe from shallow depth, while the diffusion signal emerges in deep blocks across several positions.

Table 2: GSM8K within-question AUC from all-position states at fixed depth endpoints. LLaDA and Qwen3 use generation seed 81; intervals bootstrap questions using fixed out-of-fold scores. The seed 7 replication supports the LLaDA–Qwen3 contrast, but Qwen3's own gain is inconclusive across seeds (Appendix I).

Model	First block	Deepest sampled	Depth gain	95% CI
Qwen3-8B	0.820	0.869	$+0.049$	$[+0.005, +0.094]$
LLaDA-8B	0.451	0.826	$+0.375$	$[+0.297, +0.455]$
Dream-7B	0.556	0.800	$+0.244$	$[+0.164, +0.328]$

Table 3: Cross-task within-question AUC (GSM8K→MATH). Probes are trained on GSM8K and applied directly to MATH; intervals bootstrap target questions with fixed source-trained probes. The reverse-direction contrast is reported in the text.

Model	First	Deepest	Transfer depth gain [95% CI]
Qwen3-8B	0.754	0.772	$+0.017$ $[-0.037, 0.071]$
Llama-3.1-8B	0.712	0.787	$+0.074$ $[0.009, 0.141]$
LLaDA-8B	0.517	0.736	$+0.219$ $[0.115, 0.322]$
Dream-7B	0.495	0.666	$+0.170$ $[0.079, 0.265]$

The LLaDA–Qwen3 depth-gain contrast replicates under an independent generation seed (7; the same 150 GSM8K questions, $K = 6$): $+0.284$ $[0.193, 0.370]$, versus $+0.326$ $[0.239, 0.413]$ for seed 81. Combining the seeds gives $+0.303$ $[0.241, 0.363]$. LLaDA's seed 7 gain is $+0.297$ $[0.219, 0.377]$. Qwen3's own gain remains inconclusive: its seed 7 gain is $+0.013$ $[-0.024, 0.054]$, and its combined gain is $+0.031$ $[-0.005, 0.062]$ (Appendix I).

Dream shows a similar contrast against its AR initialization family. Qwen2.5 starts at 0.850 and gains 0.093, while Dream starts at 0.556 and gains 0.244, giving a depth-gain contrast of $+0.150$ $[0.052, 0.248]$ (Appendix J). For the original four models, the contrasts also survive equal-question macro averaging and restriction to questions with mixed outcomes in both compared models (Appendix N).

Under a separate intermediate-layer protocol, Qwen3 supports stronger readouts. Mean-pooled linear AUC is 0.615 for LLaDA and 0.905 for Qwen3; an attention-pooled MLP reaches 0.739 and 0.941 (Appendix G). These are not alternative estimates of Table 2's cells.

To test transfer between numeric tasks, we train on one task and apply the same standardized readout to the other (Table 3). Both AR models transfer substantially from the first block. The diffusion models begin at chance and become transferable at later blocks, reaching 0.736 in LLaDA and 0.666 in Dream. The paired LLaDA–Qwen3 depth-gain contrast is $+0.202$ $[0.086, 0.319]$ toward MATH and $+0.198$ $[0.088, 0.314]$ in reverse.

The in-domain diffusion–AR depth-gain contrast has the same sign on MATH but remains inconclusive at $n = 150$. Transfer therefore does not establish a general architecture-level depth result. We next examine what the accessible information represents.

5 What Does the Correctness Readout Read?

To separate answer, trace, and pairing effects, we cross two factors. We split correct- and wrong-answer generations into the final numeric answer and remaining trace. This gives correct- and wrong-source traces R_c, R_w and answers A_c, A_w. Crossing them creates four inputs, with probe scores $s(R, A)$. Trace source records the outcome, not the validity of every intermediate step.

We swap only the final numeric span and check that equivalent renderings preserve the intended answer value. The cohorts contain 63 Qwen3, 79 Llama, 114 LLaDA, and 101 Dream questions. Probes are trained on the original 150 questions; each variant is scored by the fold that excluded its question. If the original six GSM8K draws lack either outcome, we make up to 48 additional attempts for these paired interventions. The resulting cohorts are not used to estimate pass rates (Appendix H).

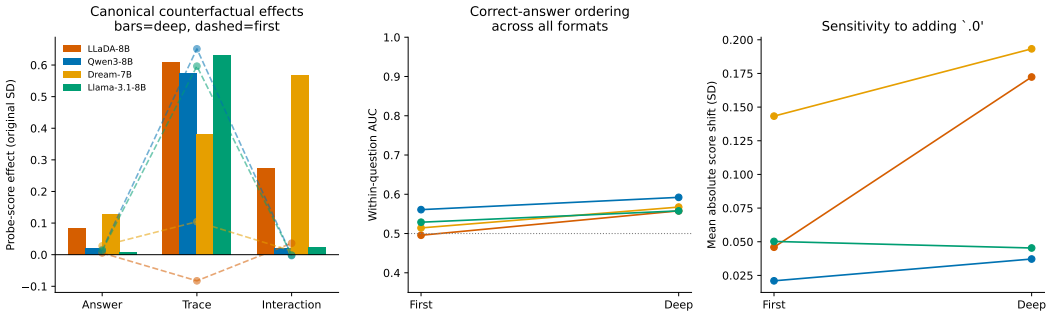


Figure 2: Counterfactual readout controls. **A:** all-position answer, trace, and interaction effects (deepest block; dashed = first block). **B:** within-question correct-answer ordering at first and deep blocks across numeric answer renderings. **C:** residual answer-format sensitivity from appending ".0".

Writing $s_{cc} = s(R_c, A_c)$ and analogously for the other pairs, we define three contrasts:

$$E_{\text{answer}} = \tfrac{1}{2}\big[(s_{cc} - s_{cw}) + (s_{wc} - s_{ww})\big], \qquad E_{\text{trace}} = \tfrac{1}{2}\big[(s_{cc} - s_{wc}) + (s_{cw} - s_{ww})\big],$$
$$E_{\text{int}} = \tfrac{1}{2}\big[(s_{cc} - s_{cw}) - (s_{wc} - s_{ww})\big]. \tag{1}$$

The main effects average changes to one factor; the interaction measures how the answer effect depends on the trace. Units are source-score standard deviations. With all-position pooling, the AR readout shows a large trace effect from the first block (≈ 0.6 SD for Qwen3 and Llama). LLaDA and Dream have no reliable first-block trace effect, but reach $+0.608$ and $+0.382$ SD at depth (Fig. 2A).

At the exact answer span (Table 4), the interaction is clearest: inserting the correct answer raises the score in a correct-source trace but lowers it in a wrong-source trace. LLaDA changes by $+1.406$ SD in R_c and -0.476 SD in R_w; Dream shows a stronger reversal. The deep readout thus depends on the answer–trace pairing.

Deep Llama also shows this interaction. The difference concerns its emergence across depth: deep-minus-first answer-span interactions are $+0.193$ for Qwen3, $+0.689$ for Llama, $+0.891$ for LLaDA, and $+1.519$ for Dream. The LLaDA–Qwen3 and Dream–Llama emergence contrasts are $+0.698$ $[0.505, 0.885]$ and $+0.830$ $[0.525, 1.124]$, respectively.

Table 4: Conditional answer effects on probe score at the deepest sampled block and exact answer positions. The first two columns replace A_w with A_c within each trace; the interaction is half their difference.

Model	$\Delta_A(R_c)$, deep/answer	$\Delta_A(R_w)$, deep/answer	E_{int}, deep/answer
Qwen3-8B	$+0.535$ $[0.348, 0.722]$	$+0.133$ $[-0.033, 0.298]$	$+0.201$ $[0.092, 0.307]$
Llama-3.1-8B	$+1.021$ $[0.816, 1.235]$	-0.399 $[-0.616, -0.169]$	$+0.710$ $[0.557, 0.871]$
LLaDA-8B	$+1.406$ $[1.205, 1.606]$	-0.476 $[-0.705, -0.246]$	$+0.941$ $[0.807, 1.069]$
Dream-7B	$+1.916$ $[1.651, 2.181]$	-1.037 $[-1.412, -0.641]$	$+1.477$ $[1.214, 1.737]$

Equivalent formats, especially leading zeros, also shift exact-answer scores. The trace effects and first-versus-deep pattern persist under all three renderings (Fig. 2). We next test pairing decoding without correctness targets and whole-state effects on prediction.

6 Late Answer States Affect Re-masked Answer Prediction

We test whether original pairings share a decodable signal and whether late answer states affect conditional scores.

Original-versus-crossed pairing is decodable at depth. We train question-grouped decoders to distinguish original pairings (R_c, A_c) and (R_w, A_w) from crossed pairings (R_c, A_w) and (R_w, A_c). The target is pairing provenance: correctness selects the source examples but is not the training label.

From the first to the deepest sampled block, all-position AUC rises from 0.501 to 0.919 in LLaDA and from 0.547 to 0.897 in Dream. In a cross-format test, training on canonical answers and testing on equivalent renderings gives AUC 0.887 and 0.886 (Table 5).

Contrast directions also become more aligned across questions. Their mean pairwise cosine rises from 0.009 to 0.390 in LLaDA and from 0.061 to 0.415 in Dream, indicating partial alignment at depth. Deep AR references also decode the pairing label. These results establish decodability, while splice-specific cues remain possible.

The signal also transfers beyond self-generated text (Appendix M): AUC reaches 0.985/0.983 on GSM8K reference rationales and 0.915/0.910 on Llama-generated traces (LLaDA/Dream). Numeric restatement nevertheless captures substantial information: 98% of correct-source GSM8K traces already state the correct value, and a text-only rule reaches 0.846/0.844. When neither trace restates its answer, only 16 questions remain per model and the decoder's intervals include chance.

A separate likelihood comparison trains decoders on all three formats. Canonical re-masked likelihood carries comparable pairing information; the decoder's advantage across all renderings partly reflects this augmentation. Table 5's cross-format test instead trains only on canonical answers.

Table 5: Original-versus-crossed pairing geometry, with pairing provenance rather than correctness as the decoder target. Cross-format decoders are trained on canonical answers and evaluated on leading-zero and decimal-equivalent answers; alignment is mean pairwise cosine between question-level contrast vectors.

Model	State	Compatibility AUC	Cross-format AUC	Contrast alignment
LLaDA-8B	first/all	0.501	0.499	0.009
LLaDA-8B	deep/all	**0.919**	**0.887**	**0.390**
Dream-7B	first/all	0.547	0.508	0.061
Dream-7B	deep/all	**0.897**	**0.886**	**0.415**

Whole late answer states alter conditional prediction. To test prediction, we place the same correct answer in R_c and R_w and re-mask it in both. At each block, forward *rescue* inserts R_c answer-prediction states into R_w; reverse *damage* does the opposite. A matched pre-answer span provides the location control.

Without patching, the correct answer is more likely under R_c by 4.118 nats in LLaDA and 6.471 in Dream. Shallow patches are null, and the effect emerges late (Fig. 3). At the deepest block, averaging rescue and damage gives $+2.704$ and $+5.148$ nats, equivalent to 66% and 80% of the mean trace gap; pre-answer effects are null or small (Table 6). Two blocks earlier, patches already recover 42% at LLaDA block 28 and 51% at Dream block 24. The deepest sites (30 of 32 and 26 of 28) are near the scoring output, where large patch fractions are partly expected; location and trace controls test specificity.

LLaDA's seed 7 patch fraction is 0.660 $[0.600, 0.713]$ on 120 paired questions (Appendix I). On short non-numeric BBH answers, deep fractions are 0.708 $[0.655, 0.755]$ in LLaDA ($n = 98$) and 0.855 $[0.840, 0.870]$ in Dream ($n = 169$). Pre-answer intervals include zero, and the effects again increase late (Appendix K).

Table 6: Whole-state patching at the deepest answer-prediction positions. The baseline row is the unpatched trace-conditioned gap; patch rows show changes in mean correct-answer log-probability. Intervals bootstrap questions. The ratio is descriptive, not a natural indirect effect.

Quantity (nats)	LLaDA-8B	Dream-7B
Baseline correct-minus-wrong-trace gap	4.118 [3.378, 4.880]	6.471 [5.473, 7.504]
Deep symmetric patch effect	+2.704 [2.056, 3.407]	+5.148 [4.327, 6.000]
Pre-answer location control	-0.114 [-0.312, 0.036]	+0.044 [0.015, 0.081]
Specific effect over control	+2.818 [2.138, 3.540]	+5.103 [4.290, 5.952]
Patch effect / mean trace gap	0.657	0.795

Dose and donor controls support specificity. We interpolate the wrong-trace answer state toward the same-question correct-trace state. The fitted rescue slope is positive for more than 90% of

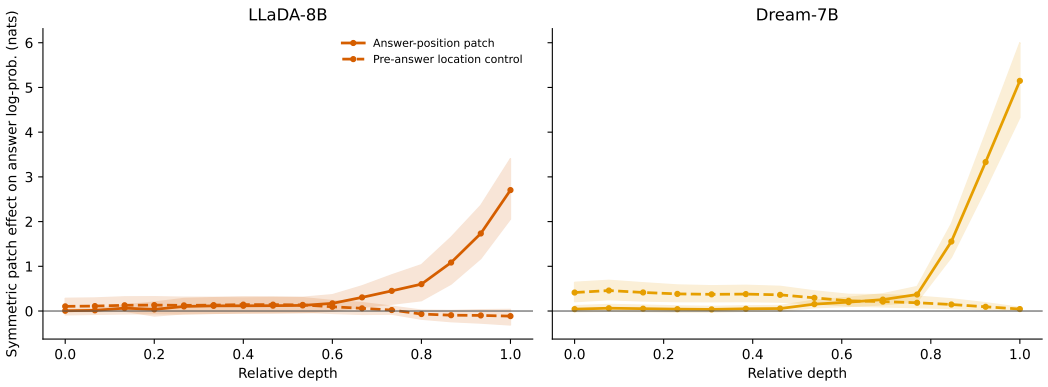


Figure 3: Causal answer-position patching across depth. Solid curves average forward rescue and reverse damage; dashed curves patch a matched pre-answer location. The answer-position effect emerges late in both models. Shading shows 95% question-bootstrap confidence intervals.

questions in each model. On 114 LLaDA and 101 Dream questions, full-dose forward rescue reaches $3.813\ [3.083, 4.642]$ and $6.302\ [5.319, 7.346]$ nats, respectively. Table 6 instead averages rescue and damage. Mean same-question rescue exceeds rescue from other-question or neuron-shuffled donors and from the matched pre-answer site. Expanding the re-masked span by one token on each side preserves the larger answer-position effect. Donor differences, intervals, and span controls appear in Appendix G.3. These single-seed checks support specificity without identifying a unique mechanism.

AR patch effects emerge earlier. Deep AR predecessor patches recover 77–100% of the trace gap, and the corrected deepest pre-answer controls are near null in all five models (Table 7). Deep localization at the answer-prediction sites is therefore shared across the models.

The difference lies earlier in the network. At 75% of depth, patches recover 6–15% of the gap in LLaDA and Dream versus 43–70% in the AR references (Table 8). The Dream–Qwen2.5 contrast is $-0.373\ [-0.448, -0.298]$ and persists on shared questions (Appendix N). AR patches target predecessor positions; diffusion patches target masked positions. This matches the tests to each predictive objective, without locating reasoning computation.

Table 7: Corrected deepest-block patch effects divided by each model's mean trace-conditioned answer-likelihood gap. The location-specific fraction subtracts the pre-answer effect from the answer-position effect. The AR controls supersede the earlier estimates. Pre-answer effects are near null in every model; the relative-depth comparison appears in Table 8.

Model	n	Answer / gap	Control / gap	Location-specific [95% CI]
LLaDA-8B	114	0.657	-0.028	0.684 [0.608, 0.758]
Dream-7B	101	0.795	0.007	0.789 [0.769, 0.807]
Qwen3-8B	60	0.768	-0.001	0.769 [0.685, 0.845]
Llama-3.1-8B	59	0.905	-0.003	0.908 [0.865, 0.951]
Qwen2.5-7B	72	1.000	0.001	0.999 [0.995, 1.003]

A shared subspace retains most of the patch effect. Whole-state patches can transfer many features. Patching only the compatibility decoder's direction yields no measurable effect (LLaDA: $0.000\ [-0.003, 0.003]$ of the whole-state effect). The visible-pair decoder also transfers poorly to masked answer slots (AUC 0.56/0.49 in LLaDA/Dream), limiting this null result.

A 64-dimensional PCA subspace of masked answer-slot $R_c - R_w$ differences, cross-fitted by question, retains $70.2\%\ [64.6\%, 76.8\%]$ in LLaDA and $83.5\%\ [81.2\%, 85.9\%]$ in Dream, versus 1.6% and 1.5% for random subspaces of the same dimension (Appendix L). The shared subspace's semantic features and complete circuit remain unresolved.

Table 8: Answer-position patch effect divided by the trace gap at fixed relative depths, using the nearest sampled block. AR effects are present from the first block; diffusion effects remain small through half depth. Intervals are available here only for the 75%-depth column.

Model	0% depth	25%	50%	75% [95% CI]
LLaDA-8B	0.00	0.02	0.03	0.15 [0.06, 0.23]
Dream-7B	0.01	0.01	0.02	0.06 [0.03, 0.08]
Qwen3-8B	0.07	0.10	0.14	0.53 [0.43, 0.62]
Llama-3.1-8B	0.15	0.20	0.45	0.70 [0.66, 0.74]
Qwen2.5-7B	0.14	0.16	0.22	0.43 [0.36, 0.50]

Trace edits change the readout and answer likelihood. We next change the trace content while holding the correct answer fixed. Changing one intermediate number lowers deep compatibility by $1.108/1.321$ SD and correct-answer likelihood by $1.172/2.052$ nats (LLaDA/Dream). Shuffling reasoning steps lowers them by $1.872/1.571$ SD and $3.529/5.663$ nats. An equivalent-format rewrite has a much smaller effect (Fig. 4).

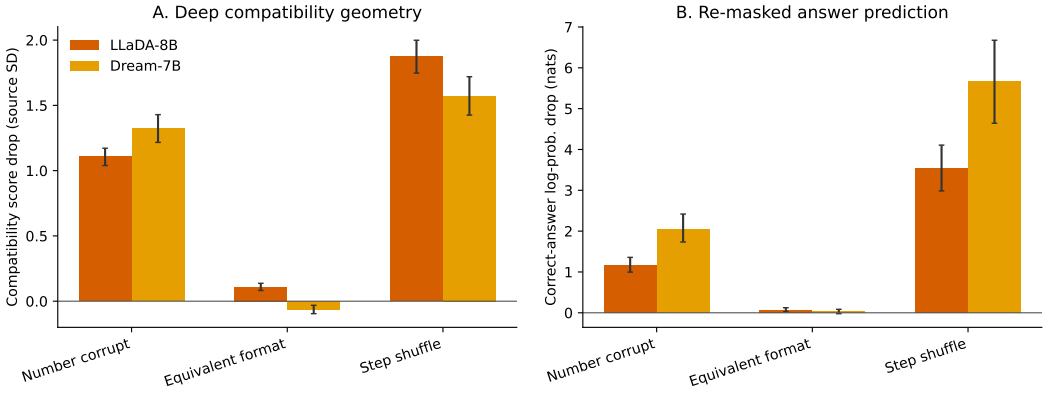


Figure 4: Holding the final answer fixed, intermediate-number corruption and step shuffling reduce both deep compatibility and re-masked correct-answer likelihood; an equivalent-format rewrite has a much smaller effect. Error bars show 95% question-bootstrap confidence intervals.

Patching the clean late *residual* state into the corrupted run recovers effects equivalent to 55%/78% of the format-adjusted number loss and 97%/92% of the ordering loss (Table 9). Attention-output patches are positive for both models and corruptions; MLP-output patches differ in sign between models. This supports an attention-output contribution in the two models, without establishing a universal component mechanism.

Table 9: Semantic-corruption patch effects at the deepest block, in nats. Each patch effect subtracts the equivalent-format edit and averages clean-to-corrupted rescue with corrupted-to-clean damage. The corresponding format-adjusted likelihood losses are 1.099/2.025 nats for number corruption and 3.456/5.636 nats for step shuffling (LLaDA/Dream). The analysis uses 113 LLaDA and 101 Dream questions.

Model	Corruption	Residual patch	Attention patch	MLP patch
LLaDA-8B	Number	0.604 [0.501, 0.714]	0.294 [0.248, 0.341]	-0.065 [-0.086, -0.044]
Dream-7B	Number	1.588 [1.294, 1.915]	0.089 [0.040, 0.140]	0.343 [0.262, 0.433]
LLaDA-8B	Step shuffle	3.350 [2.830, 3.927]	0.854 [0.724, 0.996]	-0.224 [-0.275, -0.171]
Dream-7B	Step shuffle	5.175 [4.248, 6.096]	0.860 [0.679, 1.032]	0.431 [0.330, 0.534]

7 Limitations

The LLaDA–Qwen3 GSM8K depth comparison and LLaDA decoding and patching use two generation seeds (81 and 7); Dream, Llama, Qwen2.5, MATH, factorial, component/head, and dose/donor analyses use one (Appendix H). Seed variation is therefore weakly tested. Question bootstraps keep draws and probe scores fixed, omitting variation from new seeds, probe refitting, and fold assignment. Retrospective Holm adjustment leaves 49 of 51 headline tests below $\alpha = 0.05$, using approximate p-values inferred from bootstrap intervals. This adjustment excludes other exploratory selections (Appendix N). The MATH in-domain depth contrast remains inconclusive at $n = 150$.

Shared eligible questions preserve the depth contrasts without fully matching difficulty or generation distributions. Dream's Qwen2.5 family and geometry comparison leaves other training differences, so it does not isolate the training objective.

Pairing provenance does not certify process correctness. On GSM8K, decoding substantially tracks answer restatement, and canonical re-masked likelihood carries comparable information. Format augmentation partly explains robustness to renderings; splice cues remain possible. Reference-rationale controls use GSM8K training examples.

Whole-state patches near the scoring output can transfer other features and move activations away from those normally produced by the model. Bidirectional patches, self-patches, and location controls support conditional-likelihood effects; gap fractions are not natural indirect effects. Poor masked-slot decoder transfer limits its null direction-patch result. The shared subspace does not uniquely identify a compatibility mechanism. Attention/MLP effects are non-additive, and source-token pathways remain unknown.

Causal tests cover short numeric GSM8K spans and single-token BBH option or Yes/No answers; longer free-form spans remain untested. BBH selects mixed outcomes within six samples and trims post-answer text, whereas GSM8K also uses targeted resampling for paired interventions.

8 Conclusion

Late states in LLaDA and Dream support decoding of original-versus-crossed trace–answer pairings. Transfer across formats and alignment across questions support a shared signal; on GSM8K, much of it tracks numeric consistency. Whole late answer-slot patches recover effects equivalent to 66% and 80% of the mean trace-conditioned likelihood gaps under re-masking. The LLaDA effect replicates under seed 7, and both models show late effects on short non-numeric BBH answers.

AR models share the deep localization pattern, but their patch effects emerge earlier. The tested decoder direction does not carry the patch effect, while a shared subspace retains most of it. Together, these results link late states to conditional answer scoring without establishing natural mediation by the decoded feature. The next step is to identify the subspace's semantic features and source-token paths and test longer free-form answers.

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A Measurement Details

For question i and sampled generation k, let $c_{ik} \in \{0, 1\}$ denote exact-match correctness. A readout at transformer block ℓ with pooling rule p produces

$$s_{ik}^{(\ell,p)} = f_{\ell,p}\left(h_{ik}^{(\ell)}\right),$$

where $f_{\ell,p}$ is an L2-regularized logistic regression ($C = 0.1$) over standardized states. Probes are fitted on training folds and evaluated out-of-fold, with all K generations of a question assigned to the same fold.

Pooled AUC compares every correct generation with every incorrect generation:

$$A_{\text{pool}} = \frac{\sum_{i,k:c_{ik}=1} \sum_{j,m:c_{jm}=0} \phi(s_{ik}, s_{jm})}{N_+ N_-}, \qquad \phi(a, b) = \mathbb{1}[a > b] + \tfrac{1}{2}\mathbb{1}[a = b].$$

Most comparisons have $i \neq j$. A readout that only distinguishes easy from hard questions can therefore score highly without ranking attempts to the same question.

Within-question AUC instead restricts the comparison to correct–incorrect pairs from the same question:

$$A_{\text{within}} = \frac{\sum_i \sum_{k:c_{ik}=1} \sum_{m:c_{im}=0} \phi(s_{ik}, s_{im})}{\sum_i n_i^+ n_i^-}.$$

Questions whose K generations all have the same label contribute no pairs. This micro-average weights each observed within-question pair equally. With $K = 6$, a mixed question contributes $j(6 - j)$ pairs when j attempts are correct, giving 5, 8, or 9 pairs for $j = 1$, 2, or 3. A macro-average instead takes the mean of the per-question AUCs over mixed questions, weighting each question equally. Macro endpoint values and question-bootstrap intervals for their gains appear in Appendix N.

The main *all-position* readout first averages hidden states within each of ten equal-width relative-position bins, then averages those bin means with equal bin weight before fitting one standardized logistic probe. For diffusion models, the generated region is the full 256-token canvas, including end-of-sequence filler after the answer. The *terminal-bin* readout averages the final 10% of relative generation positions, and the *exact-answer* readout averages detected answer-token positions. These pooling rules should not be conflated with selecting the best position cell in Fig. 1.

For all-position pooling, depth gain is

$$\Delta_{\text{depth}} = A_{\text{within}}^{(\ell_D,\text{all})} - A_{\text{within}}^{(\ell_0,\text{all})},$$

with both endpoints fixed in advance. Here ℓ_0 is block 0 and ℓ_D is block 30 of 32 for LLaDA, block 34 of 36 for Qwen3, and block 26 of 28 for Dream and Qwen2.5. Primary depth confidence intervals resample questions 2,000 times while keeping their generations together and holding the out-of-fold probe scores fixed. They capture evaluation-set variation, but not uncertainty from probe refitting, fold reassignment, model checkpoints, or a new generation seed. Additional controls use 2,000–5,000 question-bootstrap replicates, with the question as the statistical unit.

B Within-Question Validation

The between-question statistic in this section restricts the pooled AUC comparison to correct–incorrect pairs from *different* questions ($i \neq j$), complementing the within-question pairs defined above. Across 36 settings spanning AR and diffusion models, GSM8K, MATH, and BIG-Bench Hard [Suzgun et al., 2022], multiple temperatures, diffusion block sizes, and values of K, within-question AUC is the strongest predictor of probe-guided best-of-K selection gain. Its correlation with gain is $r = 0.762$ ($p = 6.57 \times 10^{-8}$), compared with $r = 0.525$ for the between-question signal and $r = 0.500$ for pooled AUC.

Leave-one-setting-out prediction from within-question AUC reaches $R^2 = 0.537$ with MAE 0.030, versus $R^2 = 0.171$ and MAE 0.040 for pooled AUC. A relation fitted only on AR settings predicts diffusion settings with MAE 0.023. The lower diffusion selection gains thus follow the same empirical relation in these settings.

Table 10: Correctness-readout statistics versus probe-selection gain across all 36 settings.

Readout statistic	Correlation with gain	LOO R^2	Interpretation
Within-question AUC	0.762	0.537	ranks attempts
Between-question signal	0.525	0.184	predicts difficulty
Pooled AUC	0.500	0.171	mixes both
Mean-only baseline	–	-0.058	no setting information

The distinction is visible in individual settings. Under the original intermediate-layer GSM8K protocol, Qwen3 has pooled AUC approximately 0.95 and within-question AUC approximately 0.91, whereas LLaDA has pooled AUC approximately 0.69 but within-question AUC approximately 0.62. Thus a pooled readout can appear useful while remaining weak at choosing among attempts to one prompt. This helps explain why high latent AUROC need not produce a strong best-of-N selector.

C Surface and First-Block Baselines

Seven features of the parsed answer string, with no access to model internals or the reasoning trace, already reach within-question AUC 0.780 for Qwen3, 0.634 for LLaDA, and 0.673 for Dream. Answer form therefore carries correctness information even after the question is held fixed.

The first-block hidden-state baseline differs sharply by architecture (Table 11). Qwen3's terminal-bin readout reaches 0.913, whereas LLaDA is at chance at 0.492 and Dream is much lower at 0.529. The Qwen3 result is not explained by length: restricting to correct–incorrect pairs with exactly equal generated length retains AUC 0.890. Nor is the LLaDA result caused by averaging over filler: its exact-answer first-block AUC is 0.488, and no tested shallow answer-region variant exceeds approximately 0.538. These baselines help interpret deeper readouts; their AUCs cannot be subtracted to isolate an internal signal.

Table 11: Primary-run within-question AUC for GSM8K readouts using parsed-answer features or hidden states at the indicated blocks and pooling rules. "First" is the output of transformer block 0, not the raw embedding; "terminal" is the final relative-position bin; and "exact answer" uses detected answer-token positions. Dream's terminal and exact-answer estimates independently round to 0.529.

Model	Answer only	First/all	First/terminal	First/exact answer	Deepest/all
Qwen3-8B	0.780	0.820	0.913	0.824	0.869
LLaDA-8B	0.634	0.451	0.492	0.488	0.826
Dream-7B	0.673	0.556	0.529	0.529	0.800

D Objective-Aligned Answer Scoring

For diffusion models, we jointly re-mask the generated numeric answer span and score the mean conditional log probability assigned to its original tokens. For AR models, we use the teacher-forced mean next-token log probability over the same answer span. This score requires no fitting. We treat higher likelihood as a prediction of greater correctness.

On GSM8K, the LLaDA score reaches within-question AUC 0.730 $[0.654, 0.800]$. Its interval is disjoint from Qwen3 at 0.483 $[0.449, 0.519]$ and Llama at 0.394 $[0.324, 0.460]$. Dream reaches 0.668. This is a measurement dissociation induced by differing predictive objectives, not evidence that diffusion models intrinsically know when they are correct.

Re-masking does not improve an ordinary hidden-state correctness probe. LLaDA block-0 AUC changes from 0.488 to 0.487 and its deepest AUC from 0.831 to 0.835; Dream changes from 0.529 to 0.536 at block 0. The useful conditional-likelihood scalar is therefore not evidence that masking alone localizes a better hidden-state correctness representation.

Table 12: Within-question AUC of training-free confidence scores on GSM8K. Diffusion likelihood jointly re-masks the answer span; AR likelihood scores that span left-to-right.

Model	Class	Decode-confidence AUC	Conditional answer AUC	95% CI
LLaDA-8B	diffusion	0.639	0.730	[0.654,0.800]
Dream-7B	diffusion	0.658	0.668	–
Qwen3-8B	AR	0.566	0.483	[0.449,0.519]
Llama-3.1-8B	AR	0.559	0.394	[0.324,0.460]

E Relation to Prior Readout Protocols

Yadav et al. [2026] use one generation per prompt, which supports pooled AUROC but not within-question AUC. They also report a prompt-only score of approximately 0.647 relative to a latent readout around 0.720, explicitly showing that question difficulty explains substantial signal.

We reproduce this protocol in our own harness using terminal-bin pooling, layer selection by pooled out-of-fold AUROC, and repeated draws of one of the six candidates per question. The one-sample pooled values closely track the six-sample pooled values. With one candidate per question, all comparisons are between questions, so this evaluation cannot test ordering among attempts to the same question.

Table 13: Protocol bridge. "Pooled" uses all six out-of-fold scores; "one-sample pooled" averages AUROC over random draws of one generation per question. "Same-state within" uses the pooled-selected block and pooling rule; "best all-position within" selects the block by within-question AUC.

Model	Selected block	Pooled	One-sample pooled	Same-state within	Best all-position within
LLaDA-8B	30	0.686	0.684 ± 0.032	0.776	0.826
Qwen3-8B	26	0.967	0.967 ± 0.012	0.930	0.900
Dream-7B	20	0.777	0.776 ± 0.025	0.706	0.800

The bridge shows why pooled AUC alone is hard to interpret. First, Qwen3's block-0 terminal state already has pooled AUROC 0.939 and within-question AUC 0.913. Second, pooled and within-question layer selection can disagree: Dream's pooled-selected state has within-question AUC 0.706, whereas deepest all-position pooling reaches 0.800. Pooled AUROC alone cannot identify whether a readout uses question difficulty, answer form, or internal information that distinguishes attempts.

F Mechanistic Tests and Negative Results

We test two possible explanations for the depth pattern. First, varying parallel decoding at fixed compute gives LLaDA within-question AUC 0.664/0.784/0.720/0.704 for block sizes 256/64/32/8. The relation to AR-likeness is inconclusive ($r = 0.22$, $p = 0.78$), so this test does not explain the pattern.

Second, we reorder states by the denoising step at which tokens are committed. The slope is positive in LLaDA ($+0.125$) and negative in Dream (-0.022), providing no common aggregation pattern across the two models. Re-masking also leaves hidden-state readout performance largely unchanged (Appendix D).

These representational findings yield little practical selection gain. On numeric diffusion tasks, combining probe score, agreement, and confidence has an observed advantage of 0.011 over majority voting ($p = 0.14$). Probe-guided allocation matches random. The evidence therefore supports interpreting the readouts, while improved selection remains unestablished.

G Additional Results

G.1 Reader-capacity ladder on a separate protocol

These comparisons use an earlier intermediate-layer protocol. Their AUCs are therefore not alternative estimates of the fixed-endpoint GSM8K results in Table 2. Under identical folds, greater reader

capacity can improve LLaDA's scores without closing the gap to Qwen3 (Tables 14, 15). Per-token states and learned pooling help, but LLaDA's best reader remains below Qwen3's weakest. The gap therefore cannot be explained solely by the capacity of a linear probe.

Table 14: Mean-pooled reader capacity on the earlier intermediate-layer protocol. Values are within-question AUC.

Mean-pooled reader	Qwen3-8B	LLaDA-8B	Dream-7B
Linear, one layer	0.905	0.615	0.766
Linear, all-layer concat.	0.903	0.623	0.768
MLP, all-layer concat.	0.881	0.652	0.685
Gradient boosting	0.838	0.639	0.719
Linear, per-question centered	0.892	0.613	0.728

Table 15: Per-token reader capacity on the earlier intermediate-layer protocol. Values are within-question AUC.

Per-token reader	Qwen3-8B	LLaDA-8B
Mean pool + linear	0.899	0.658
Mean pool + MLP	0.901	0.666
Max pool + linear	0.947	0.734
Attention pool + MLP	0.941	0.739
Last token + linear	0.956	0.657

G.2 Further controls

Prompt-only difficulty accounts for most of LLaDA's between-question signal: the best prompt-only correlation is 0.412 versus 0.438 for generations, compared with 0.651 versus 0.862 in AR models. Under difficulty matching, within-question AUC is 0.903 for Qwen3 and 0.625 for LLaDA. LLaDA is weakest when only one of six attempts is correct: AUC rises from 0.489 at $j{=}1$ to 0.685 at $j{=}5$.

Selection performance also depends on the task. Majority voting beats the original LLaDA probe on numeric GSM8K (0.670 versus 0.625), while the probe wins on all four examined BIG-Bench Hard settings.

About 2.5%/2.6% of AR/diffusion generations lack a parseable answer. Restricting evaluation to questions with all answers parseable preserves the probe's advantage over decode confidence.

G.3 Dose, donor, and span controls

On 114 LLaDA and 101 Dream paired questions, interpolating the wrong-trace answer state toward the same-question correct-trace state raises mean correct-answer log-likelihood from zero rescue at dose $\alpha = 0$ to 3.813 [3.083, 4.642] nats in LLaDA and 6.302 [5.319, 7.346] nats in Dream at $\alpha = 1$. The fitted rescue slope is positive for more than 90% of questions in each model. At full dose, same-question donor rescue exceeds the three controls by the mean amounts in Table 16.

Table 16: Full-dose donor controls on 114 LLaDA and 101 Dream questions. Entries are same-question rescue minus control rescue (nats), with bracketed confidence intervals.

Control donor or site	LLaDA-8B	Dream-7B
Other-question donor	5.170 [4.590, 5.778]	5.392 [4.752, 6.017]
Neuron-shuffled donor	3.513 [2.870, 4.193]	5.681 [4.833, 6.556]
Matched pre-answer site	4.100 [3.285, 4.979]	6.219 [5.245, 7.234]

Expanding the re-masked answer by one neighboring token on each side gives deepest answer-position effects equivalent to 61% (LLaDA) and 75% (Dream) of the corresponding trace gaps. The

location-specific effects are 2.272 [1.261, 3.314] and 3.319 [2.649, 4.017] nats, respectively, while matched pre-answer controls are $-0.093\,[-0.272, 0.041]$ and $0.050\,[0.018, 0.090]$.

H Reproducibility

Original and additional generation and analysis were run in Kaggle GPU notebooks (NVIDIA T4/P100), using checkpoints from the Hugging Face hub. The primary GSM8K and MATH runs cover 150 questions with $K = 6$ generations for all four models. The primary LLaDA and Qwen3 depth results use seed 81. An independent seed 7 run is complete on the same 150 GSM8K questions for these two models (Appendix I). LLaDA decoding and whole-state patching also replicate under seed 7; Dream, Llama, Qwen2.5, MATH, the factorial interaction, and component/head analyses remain single-seed.

The crossed cohorts contain 114 LLaDA, 101 Dream, 63 Qwen3, and 79 Llama questions, with all twelve variants per question passing numeric checks. Diffusion activation patching and donor-specificity analyses use 114 LLaDA and 101 Dream questions. The semantic-component analysis uses 113 LLaDA and 101 Dream questions after excluding one LLaDA question without a valid semantic corruption. The objective-matched AR location controls use 60 Qwen3, 59 Llama, and 72 Qwen2.5 questions.

All-correct and all-wrong prompts are resampled until the missing outcome appears, with up to 48 additional attempts. This targeted sampling is used only for the GSM8K paired interventions and never for pass-rate estimation. BBH uses mixed outcomes within the original six samples without targeted resampling.

Exact self-patches reproduce the unpatched score to machine precision. Generators checkpoint their progress, and each analysis truncates to the completion marker, so no partial artifact is treated as complete.

The manuscript does not supply executable scripts, exact run manifests, checkpoint revision hashes, or derived result arrays. These omissions limit independent reproduction of the reported numerical estimates.

I Independent Generation Seed

Seed 7 uses the same 150 GSM8K questions, $K = 6$, and generation settings as the primary seed 81 LLaDA and Qwen3 runs. The seed 7 arrays combine shards with complete question coverage and no overlap. Estimates were recomputed from the saved hidden-state arrays using the same folds, probe, and question-bootstrap procedure as the primary analysis. Table 17 reports all-position within-question AUCs and gains with the first and deepest endpoints held fixed: blocks 0 and 30 for LLaDA, and blocks 0 and 34 for Qwen3. Qwen3's post hoc best block is not substituted for the pre-specified endpoint. Per-sample seed identifiers are computed as $100003\,\sigma + 131i + k$, where σ is the generation seed, i the question index, and k the sample index.

The LLaDA–Qwen3 contrast is paired by question. Combining the seeds adds generation draws on the same 150 questions, not independent questions. The pooled estimate sums each seed's within-question pair contributions per question, and the bootstrap retains both seeds for each resampled question. The 95% question-bootstrap intervals condition on the observed seeds and fixed out-of-fold scores; they do not estimate variation over new seeds.

The seed-81-minus-seed-7 gain differences are $+0.079\ \ [-0.031, 0.182]$ for LLaDA and $+0.036\ [-0.017, 0.089]$ for Qwen3. These intervals do not detect a gain difference between the two observed seeds and do not establish equivalence. Qwen3's first-block AUC rises from 0.820 to 0.847 while its deepest-block AUC changes little (0.869 to 0.859), leaving its own depth gain small and inconclusive in seed 7 and in the combined estimate.

LLaDA's late increase also recurs: at the sampled blocks in Table 18, micro-AUC remains at or below 0.641 through block 26 in both seeds, then rises at blocks 28 and 30. Equal-question macro averages show the same endpoint pattern; their gain intervals are reported in Appendix N. Per-question accuracy correlates across seeds at $r = 0.853$ for LLaDA and $r = 0.890$ for Qwen3.

Table 17: Fixed-endpoint GSM8K depth replication. Brackets show 95% question-bootstrap confidence intervals. Combined endpoint AUCs are descriptive; no intervals are reported for them. Rounding can make subtraction of displayed endpoints differ by 0.001 from reported gains.

Readout or contrast	Seed 81	Seed 7	Combined
LLaDA, block 0	0.451 $[0.383, 0.524]$	0.502 $[0.434, 0.566]$	0.477
LLaDA, block 30	0.826 $[0.771, 0.877]$	0.798 $[0.737, 0.858]$	0.812
LLaDA depth gain	$+0.375$ $[0.297, 0.455]$	$+0.297$ $[0.219, 0.377]$	$+0.334$ $[0.276, 0.391]$
Qwen3, block 0	0.820 $[0.753, 0.878]$	0.847 $[0.796, 0.897]$	0.833
Qwen3, block 34	0.869 $[0.813, 0.918]$	0.859 $[0.811, 0.905]$	0.864
Qwen3 depth gain	$+0.049$ $[0.005, 0.094]$	$+0.013$ $[-0.024, 0.054]$	$+0.031$ $[-0.005, 0.062]$
LLaDA–Qwen3 gain	$+0.326$ $[0.239, 0.413]$	$+0.284$ $[0.193, 0.370]$	$+0.303$ $[0.241, 0.363]$

Table 18: LLaDA all-position AUC at sampled GSM8K blocks. Micro averages weight within-question pairs equally; macro averages weight mixed questions equally. These per-layer values are descriptive. Macro depth-gain intervals appear in Appendix N.

Seed / average	0	8	16	20	24	26	28	30
81 / micro	0.451	0.522	0.577	0.609	0.577	0.607	0.719	0.826
7 / micro	0.502	0.551	0.603	0.597	0.625	0.641	0.739	0.798
81 / macro	0.443	0.511	0.572	0.614	0.579	0.602	0.723	0.832
7 / macro	0.497	0.555	0.601	0.602	0.638	0.647	0.747	0.806

J Dream Versus Its AR Initialization Family

Dream was initialized from Qwen2.5-7B. We repeat the depth sweep, original-versus-crossed decoding, and AR patching with Qwen2.5-7B-Instruct, which matches Dream's 28 blocks and hidden width of 3584. On the same 150 GSM8K questions, Qwen2.5 accuracy is 0.558 and pass@6 is 0.793, with 74 mixed-outcome questions; the intervention cohort contains 72 questions. This is a comparison within the initialization family, not an isolated training-objective ablation.

K Causal Patching on Non-Numeric Answers

We repeat re-masked answer patching on 358 questions from eight BBH tasks: logical deduction, shuffled-object tracking, date understanding, colored objects, temporal sequences, penguins, navigate, and web of lies. Answers are multiple-choice letters or Yes/No. Each model samples six generations per question ($T = 0.9$, 128 denoising steps, 256-token canvas). Questions with at least one correct and one wrong parseable answer form the cohort, without targeted resampling.

Table 19: Independent-seed LLaDA decoding and whole-state patching. Brackets show 95% question-bootstrap intervals; the AUC interval applies to the deep endpoint. At blocks 24, 26, 28, and 30, patch/gap fractions are 0.15, 0.28, 0.44, and 0.66 in seed 7 versus 0.15, 0.26, 0.42, and 0.66 in seed 81. Eligibility yields different paired cohorts across seeds.

LLaDA-8B, GSM8K	Seed 81	Seed 7
Paired questions	114	120
Trace gap (nats)	4.118 [3.378, 4.880]	3.935 [3.256, 4.648]
Deep symmetric patch (nats)	2.704 [2.056, 3.407]	2.598 [2.023, 3.232]
Pre-answer control (nats)	-0.114 $[-0.312, 0.036]$	-0.038 $[-0.201, 0.085]$
Patch / gap	0.657	0.660 [0.600, 0.713]
Pairing AUC, first / deep	0.501 / 0.919	0.500 / 0.918 [0.894, 0.941]

Table 20: Dream and its AR initialization family. The Dream–Qwen2.5 depth-gain contrast is $+0.150$ $[0.052, 0.248]$; the 75%-depth patch fraction contrast is -0.373 $[-0.448, -0.298]$. Qwen2.5 has a positive depth gain, while Dream's gain is larger from a lower shallow floor. The pairing-AUC interval applies to Qwen2.5's deep endpoint.

Quantity	Qwen2.5-7B (AR)	Dream-7B (diffusion)
Within-question AUC, block 0	0.850 [0.794, 0.897]	0.556 [0.467, 0.642]
Within-question AUC, block 26	0.943 [0.906, 0.972]	0.800 [0.731, 0.860]
Depth gain	$+0.093$ $[0.051, 0.140]$	$+0.244$ $[0.164, 0.328]$
Pairing AUC, first / deep	0.585 / 0.896 [0.870, 0.920]	0.547 / 0.897
Deepest patch / gap	1.000 ($n = 72$)	0.795 ($n = 101$)
75%-depth patch / gap	0.43 [0.36, 0.50]	0.06 [0.03, 0.08]
Deepest pre-answer control / gap	0.001	0.007

The answer is the option after the last "####" marker or, if that marker is absent, after the model's "the answer is" or "final answer" statement. In both traces, this span is replaced by the gold answer and subsequent text is removed. Only the option letter or Yes/No token is masked and scored. Every block is patched, and self-patches reproduce the unpatched score exactly. This trimming and eligibility policy differs from GSM8K and should be considered when comparing the fractions.

Table 21: Non-numeric BBH answer patching at LLaDA block 30 and Dream block 26. Brackets show 95% question-bootstrap intervals; pairing-AUC intervals apply to the deep endpoints. Answer-type fractions are descriptive. Patch/gap rises from 0.18 to 0.36 to 0.51 at LLaDA blocks 24, 26, and 28, and from 0.24 to 0.41 to 0.64 at Dream blocks 20, 22, and 24. Pre-answer intervals include zero.

Quantity	LLaDA-8B	Dream-7B
Paired questions	98	169
Trace gap (nats)	6.752 [5.690, 7.808]	3.590 [3.105, 4.066]
Deep symmetric patch (nats)	4.778 [3.832, 5.753]	3.070 [2.659, 3.481]
Pre-answer control (nats)	0.004 $[-0.001, 0.009]$	-0.025 $[-0.143, 0.092]$
Patch / gap	0.708 [0.655, 0.755]	0.855 [0.840, 0.870]
Multiple-choice letters	0.603 ($n = 69$)	0.837 ($n = 114$)
Yes/No	0.956 ($n = 29$)	0.923 ($n = 55$)
Pairing AUC, first / deep	0.510 / 0.882 [0.844, 0.917]	0.531 / 0.845 [0.814, 0.877]

L Directional and Subspace Patching

At the deepest sampled block, we patch a projection of the masked answer-slot state in both rescue and damage directions:

$$h \leftarrow h + P(h_{\text{donor}} - h).$$

Here P projects onto the tested subspace. Subspaces are cross-fitted by question using five folds; learned PCA bases use $R_c - R_w$ masked-slot differences from training questions. Whole-state patching reproduces Table 6.

The visible-pair compatibility decoder captures only 0.2%/0.1% of masked-slot difference energy and decodes pairing poorly from these states (AUC 0.56/0.49 in LLaDA/Dream). Mean pairwise cosine of question differences is $+0.05/+0.03$. These observations do not prove the absence of every shared direction; they limit the tested rank-one explanations. At dimension 64, the retained LLaDA fraction rises from 0.21 at block 22 to 0.33 at block 26, 0.50 at block 28, and 0.70 at block 30.

The masked-slot decoder has an additional identifiability limitation: for a fixed trace, digit-per-token tokenization can make the re-masked inputs identical when the two answers have equal length. A visible-answer direction variant also yields a null effect.

The subspace result supports a shared component of the intervention effect, but does not identify a unique semantic compatibility feature, natural mediation, or a complete circuit.

Table 22: Fraction of the whole-state patch effect retained by each subspace. Brackets are 95% question-bootstrap intervals where available; rank-one mean and random-subspace values are descriptive. Tested rank-one directions are insufficient, while the cross-fitted 64-dimensional PCA subspace preserves most of the effect.

Subspace	Dimension	LLaDA ($n = 114$)	Dream ($n = 101$)
Compatibility decoder	1	$0.000\ [-0.003, 0.003]$	$0.001\ [-0.000, 0.003]$
Mean $R_c - R_w$ difference	1	-0.039	0.031
PCA of $R_c - R_w$	8	0.346 [0.286, 0.415]	0.440 [0.394, 0.486]
PCA of $R_c - R_w$	16	0.564 [0.513, 0.623]	0.670 [0.631, 0.709]
PCA of $R_c - R_w$	64	0.702 [0.646, 0.768]	0.835 [0.812, 0.859]
Random subspace	64	0.016	0.015

M Transfer, Likelihood, and Answer-Restatement Controls

Transfer beyond self-generated text. Decoders trained on each model's own original and crossed pairs are cross-fitted by question and evaluated on text from other sources. GSM8K reference rationales come from the training split; the 150 questions are selected from its first 640 examples. This is a representational transfer control, not a held-out benchmark accuracy result.

Table 23: Decoder transfer to reference and other-model traces, with 95% question-bootstrap intervals where available. Successful transfer shows that the readout is not restricted to text generated by the model whose states are probed; numeric and lexical consistency cues remain possible.

Trace source and pairing comparison	LLaDA	Dream
Own traces	0.919	0.897
Reference rationale: gold vs. wrong answer	0.985 [0.971, 0.997]	0.983 [0.968, 0.994]
Qwen3 traces ($n = 63$)	0.817 [0.777, 0.856]	0.757 [0.710, 0.802]
Llama traces ($n = 79$)	0.915 [0.889, 0.940]	0.910 [0.884, 0.934]

Decoder versus re-masked likelihood. Canonical-format likelihood has slightly higher point AUC than the decoder (Table 24). Adding the decoder to likelihood gives $+0.018\ [0.000, 0.037]$ in LLaDA and $+0.005\ [-0.015, 0.025]$ in Dream. Across three renderings, likelihood weakens on equivalent rewrites, while the decoder has been trained on all three formats. Its rendering robustness is therefore partly a property of the training design.

Table 24: Original-versus-crossed pairing AUC by answer format and score, using a different target from the correctness AUC in Table 12. These point estimates are descriptive; "–" means not reported. Rewriting originals as well as crossed pairs leaves decoder performance similar, so it does not require originals to be uniquely unedited.

	LLaDA		Dream	
Answer formats	Decoder	Likelihood	Decoder	Likelihood
Canonical	0.925	0.936	0.908	0.923
All three renderings	0.919	0.771	0.897	0.671
Every variant rewritten	0.916	–	0.897	–

Answer restatement. In 98% of correct-source GSM8K traces, the correct value appears before the final answer. A text-only rule asking whether the final value appears earlier in the trace gives original-versus-crossed AUC 0.846 in LLaDA and 0.844 in Dream, versus 0.587 in Qwen3 and 0.706 in Llama. Table 25 conditions decoder evaluation on this cue.

These controls show a substantial relationship with numeric restatement, but the binary rule is not a complete explanation: decoder AUC remains high when both traces restate their answers. The small neither-restated and against-label subsets limit conclusions about cue-independent compatibility.

Table 25: Decoder AUC conditional on the answer-restatement cue. The neither-restated diffusion subsets contain 16 questions each and have wide 95% question-bootstrap intervals. Other values are descriptive where no interval is supplied.

Cue relation to label	LLaDA	Dream	Qwen3	Llama
Aligned	0.943	0.960	0.899	0.835
Tied: both restated	0.960	0.863	0.822	0.861
Tied: neither restated	0.644 [0.391, 0.889]	0.542 [0.357, 0.719]	0.783	0.903
Against label	0.500 ($n_q = 11$)	0.097 ($n_q = 8$)	0.921	0.815

In the latter Dream subset, the decoder favors the crossed pairing when the wrong-source trace already contains the correct value; this is consistent with numeric agreement rather than generation provenance alone.

N Macro-Averaging, Shared-Question Comparisons, and Multiplicity

Equal-question macro AUC. Giving each mixed question equal weight preserves the large diffusion depth gains (Table 26). The intervals below are question-bootstrap estimates for macro gains; per-layer point values in Table 18 remain descriptive.

Table 26: Macro within-question AUC at fixed depth endpoints. Gain intervals resample questions; endpoint intervals are not supplied. Rounding can make displayed endpoint subtraction differ by 0.001.

Model	Seed	First	Deepest	Macro gain [95% CI]
LLaDA-8B	81	0.443	0.832	$+0.389$ $[0.310, 0.468]$
Dream-7B	81	0.556	0.807	$+0.251$ $[0.166, 0.336]$
Qwen3-8B	81	0.816	0.871	$+0.056$ $[0.007, 0.107]$
Llama-3.1-8B	81	0.756	0.774	$+0.018$ $[-0.034, 0.068]$
LLaDA-8B	7	0.497	0.806	$+0.309$ $[0.233, 0.383]$
Qwen3-8B	7	0.841	0.858	$+0.016$ $[-0.024, 0.061]$

Shared eligible questions. Depth comparisons restricted to questions with mixed outcomes for both models retain positive contrasts. Restricting the 75%-depth patch comparison to shared questions also preserves its direction (Table 27). These restrictions hold question identity fixed but do not match every dimension of difficulty or each model's generation distribution.

Corrected AR controls. Rerunning the current AR patching script on identical seed 81 inputs reproduces the baseline gaps and answer-position effects exactly but changes the pre-answer controls. Table 7 uses the corrected values. The deepest location-specific contrasts are -0.085 $[-0.191, 0.027]$ for LLaDA–Qwen3, -0.119 $[-0.167, -0.073]$ for Dream–Llama, and -0.211 $[-0.230, -0.192]$ for Dream–Qwen2.5. The earlier claim of greater diffusion-specific late location concentration is withdrawn. The Qwen3 seed 7 pre-answer check is -0.032 $[-0.047, -0.018]$ nats (-0.2% of the trace gap), providing an additional small-effect control; it is not a full second-seed causal replication for all AR comparisons.

Multiplicity sensitivity analysis. A Holm adjustment over 51 headline tests, using two-sided normal-approximation p-values inferred from the reported bootstrap intervals, leaves 49 significant at $\alpha = 0.05$. Qwen3's depth gain and Llama's transfer gain do not survive. This is a retrospective sensitivity analysis of the stated test family: normal approximation to bootstrap intervals and selection of headline tests limit its interpretation, and the adjustment does not cover all exploratory layers, formats, or subgroups. The corrected AR-control analysis supersedes the older location tests in this calculation.

Table 27: Shared-question sensitivity analyses. Panel (a) compares correctness AUC gains; panel (b) compares patch effects divided by each model's trace gap. Brackets show 95% question-bootstrap intervals.

(a) Depth gains on jointly mixed questions

Contrast	Shared n	Depth-gain contrast [95% CI]
LLaDA–Qwen3, seed 81	41	$+0.429$ $[0.307, 0.549]$
LLaDA–Qwen3, seed 7	39	$+0.343$ $[0.225, 0.468]$
LLaDA–Llama	54	$+0.323$ $[0.209, 0.447]$
Dream–Qwen3	34	$+0.270$ $[0.150, 0.389]$
Dream–Llama	40	$+0.231$ $[0.109, 0.351]$

(b) Answer-position patch contrasts at 75% depth

Contrast	All eligible questions	Shared questions
LLaDA–Qwen3	-0.380 $[-0.505, -0.251]$	-0.479 $[-0.617, -0.343]$ ($n = 50$)
Dream–Llama	-0.641 $[-0.690, -0.594]$	-0.609 $[-0.664, -0.557]$ ($n = 45$)
Dream–Qwen2.5	-0.373 $[-0.448, -0.298]$	-0.373 $[-0.463, -0.284]$ ($n = 56$)

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- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
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